

Walkthrough

Why this code was created?

To download videos from Mediathek using simple **fetch api** for url, **jsdom** to find to navigate and search for the elements: that give the episode names. that directs to the mp4 file, and as well the **stream** to download the mp4 file.

In this case I used this series **Tatort Münster** (wiki page: https://en.wikipedia.org/wiki/Thiel_and_Boerne)

How this code works?

- 1) we search for a website page using fetch api where we extract the episode names using jsdom.
- 2) which will then will be used in other fetch api; this fetch api (url) will have the format of:
Mediathek domain/search + episodes names as query params
- 3) the episodes that are found will be then be downloaded using createWriteStream

In depth walktrough:

1) Retrieve episode names from this url page

<https://tatort-fans.de/category/stadt-archiv/tatort-muenster/>



h2 class="teaser-title";

inside h2 is <a href="some link regarding the info about the episode"

class="class="text-decoration-none stretched-link"

Tatort Folge 1299: Fiderallala>

The code for this part looks like this:

```
const myURL = "https://tatort-fans.de/category/stadt-archiv/tatort-muenster/";

function getAllTitles(titles) {
  return titles.map((title) => title.children[0].text.split(":")[1].trim());
}

async function gimmeTitles() {
  try {
    const response = await fetch(myURL);
    if (response.ok) {
      const data = await response.text();
      const dom = new JSDOM(data);
      const document = dom.window.document;

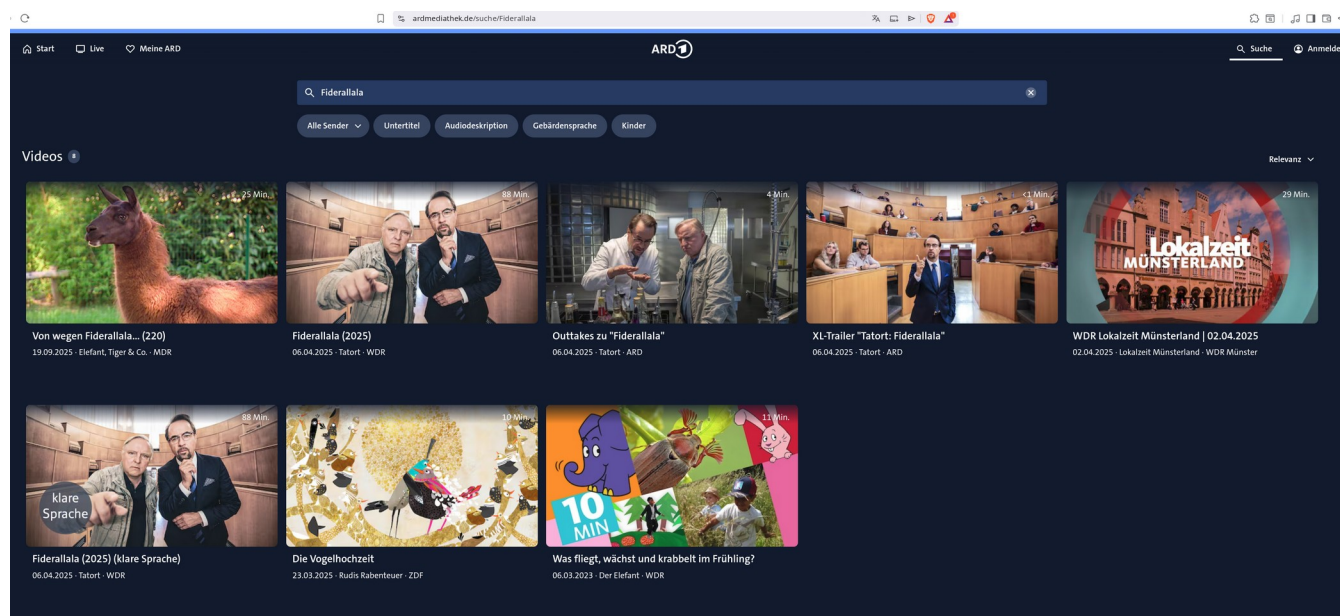
      const titles = Array.from(document.querySelectorAll(".teaser-title"));

      const listTitles = getAllTitles(titles);
      return listTitles;
    } else {
      console.log("Failed :(");
      return [];
    }
  } catch (err) {
    console.error(err);
    return [];
  }
}
```

- the asynchronous function `gimmeTitles()` fetches the url and from the `response.text()` we take the data.
- the data is being parsed into JSDOM so we can retrieve the episode names that are present in class `.teaser-title` using `querySelectorAll`
- this will retrieve the episode names in the format of `[Tatort Folge _number: _nameOfEpisode]`
- but before finishing, we call function `getAllTitles` to split the episodes names into two:

```
[0] Tatort Folge _number  split(":") [1] _nameOfEpisode
```

2) Search for the episode in Mediathek



Before seeing the code, let's see how would a normal person would search for; We have one episode name, we searched it, and we want 3 versions for this type episode.

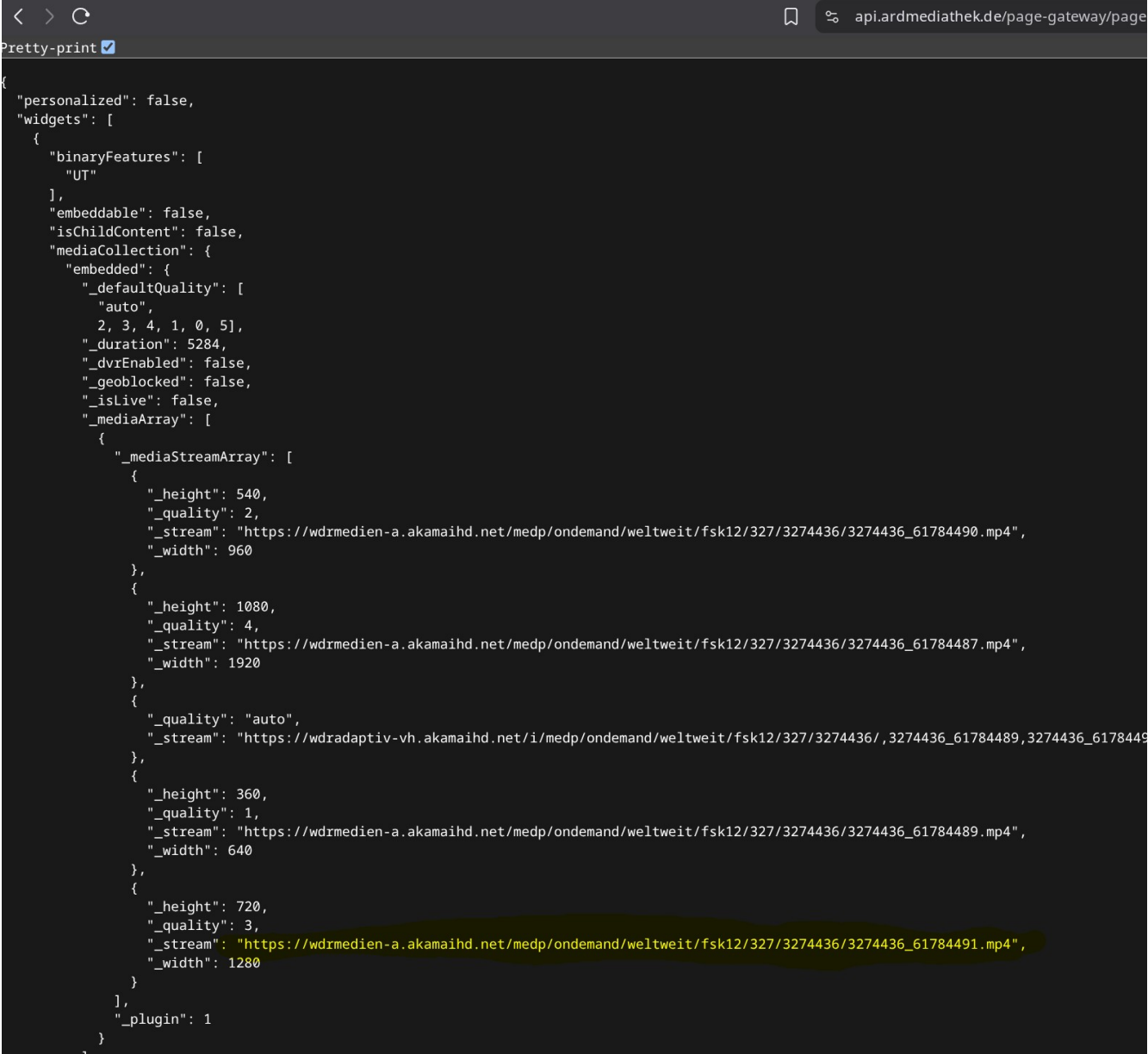
I chose to click on the one that has "klare Sprache" since my german listening is not so good, and I would like to have a better comprehension.

Let's keep in mind that the code will search for the episode with that name + "klare Sprache" and exclude the Outtakes.

But how it is done exactly?

Well if type ourselves the name of the episode (in this case Fiderallala klare Sprache), we will get the episode. WHAT IS THE MOST IMPORTANT THING IS THAT WHEN INSPECTING AND GOING TO THE NETWORK THE LAST API FROM THIS WEBSITE IS SOMETHING LIKE THIS:

we'll get a JSON Object as Reponse:



```
{
  "personalized": false,
  "widgets": [
    {
      "binaryFeatures": [
        "UT"
      ],
      "embeddable": false,
      "isChildContent": false,
      "mediaCollection": {
        "embedded": {
          "_defaultQuality": [
            "auto",
            2, 3, 4, 1, 0, 5],
          "_duration": 5284,
          "_dvrEnabled": false,
          "_geoblocked": false,
          "_isLive": false,
          "_mediaArray": [
            {
              "_mediaStreamArray": [
                {
                  "_height": 540,
                  "_quality": 2,
                  "_stream": "https://wdrmedien-a.akamaihd.net/medp/ondemand/weltweit/fsk12/327/3274436/3274436_61784490.mp4",
                  "_width": 960
                },
                {
                  "_height": 1080,
                  "_quality": 4,
                  "_stream": "https://wdrmedien-a.akamaihd.net/medp/ondemand/weltweit/fsk12/327/3274436/3274436_61784487.mp4",
                  "_width": 1920
                },
                {
                  "_quality": "auto",
                  "_stream": "https://wdradaptiv-vh.akamaihd.net/i/medp/ondemand/weltweit/fsk12/327/3274436/,3274436_61784489,3274436_61784491"
                },
                {
                  "_height": 360,
                  "_quality": 1,
                  "_stream": "https://wdrmedien-a.akamaihd.net/medp/ondemand/weltweit/fsk12/327/3274436/3274436_61784489.mp4",
                  "_width": 640
                },
                {
                  "_height": 720,
                  "_quality": 3,
                  "_stream": "https://wdrmedien-a.akamaihd.net/medp/ondemand/weltweit/fsk12/327/3274436/3274436_61784491.mp4",
                  "_width": 1280
                }
              ]
            },
            1,
            "_plugin": 1
          ]
        }
      }
    }
  ]
}
```

Inside mediaStreamArray there 4 types of video quality, we take the last & highest one of the video and if we copy the stream, we get the video

